

Dynamic ARP Inspection

- Topics covered:
 - What Is Dynamic ARP Inspection
 - How Does It Work?
 - What Attacks Does It Prevents?
 - DAI Configuration

What Is Dynamic ARP Inspection?

Introduction and ARP Review

ARP (Address Resolution Protocol) is used to learn the MAC address of a device when its IP address is already known.

A PC uses ARP to:

1. Learn the MAC address of its default gateway when communicating with external networks.
2. Learn the MAC address of other devices on the local network.

Typically, ARP uses a two-message exchange:

1. ARP Request
2. ARP Reply

Example Scenario:

PC1 wants to send a DNS query to **8.8.8.8**.

PC1 determines that 8.8.8.8 is outside its local network, so it must send the frame to its default gateway (**192.168.1.1**).

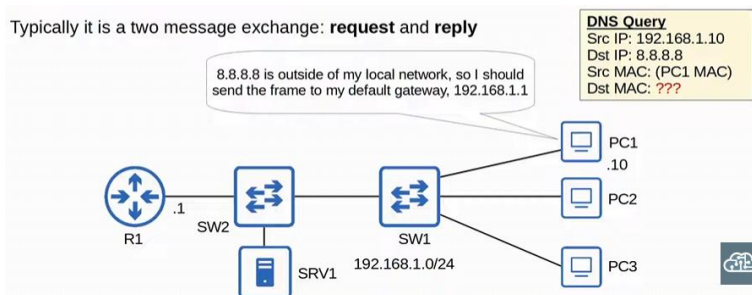
However, PC1 does not know the MAC address of the default gateway (R1).

PC1 needs R1's MAC address before sending the DNS query.

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Network Topology

Typically it is a two message exchange: **request** and **reply**



ARP Request

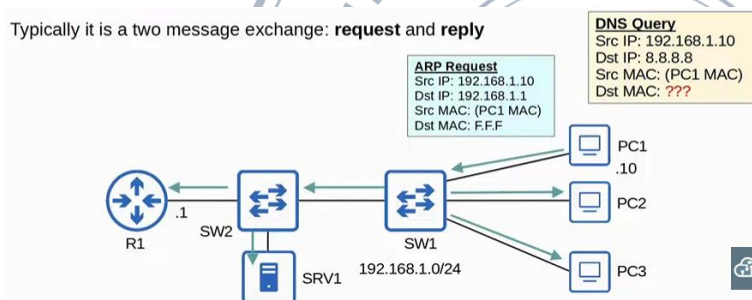
PC1 broadcasts an ARP Request asking:

Who has IP address 192.168.1.1?

Because the destination MAC is the broadcast MAC address, every device in the LAN receives the request.

ARP Request Flow

Typically it is a two message exchange: **request** and **reply**



ARP Request in Wireshark

```
> Frame 99: 60 bytes on wire (480 bits), 60 bytes captured (480 bits) on interface 0
  ✓ Ethernet II, Src: 0c:29:2f:90:91:00 (0c:29:2f:90:91:00), Dst: Broadcast (ff:ff:ff:ff:ff:ff)
    > Destination: Broadcast (ff:ff:ff:ff:ff:ff)
    > Source: 0c:29:2f:90:91:00 (0c:29:2f:90:91:00)
      Type: ARP (0x0806)
      Padding: 00000000000000000000000000000000
  ✓ Address Resolution Protocol (request)
    Hardware type: Ethernet (1)
    Protocol type: IPv4 (0x0800)
    Hardware size: 6
    Protocol size: 4
    Opcode: request (1)
    Sender MAC address: 0c:29:2f:90:91:00 (0c:29:2f:90:91:00)
    Sender IP address: 192.168.1.10
    Target MAC address: 00:00:00_00:00:00 (00:00:00:00:00:00)
    Target IP address: 192.168.1.1
```

Ethernet Header-Ethernet II

Destination: Broadcast (ff:ff:ff:ff:ff:ff)

Source: 0c:29:2f:90:91:00

Type: ARP (0x0806)

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ARP Header-Address Resolution Protocol (request)

Sender MAC address: 0c:29:2f:90:91:00

Sender IP address: 192.168.1.10

Target MAC address: 00:00:00:00:00:00

Target IP address: 192.168.1.1

Important Note Although the ARP message is encapsulated in an Ethernet frame, it does **not** contain an IP header.

Therefore, when referring to:

- Source IP → Sender IP Address field
- Destination IP → Target IP Address field

these values come from the ARP message itself.

ARP messages are only broadcast within the local network and are not forwarded by routers to external networks, so an IP header is unnecessary.

Keep these ARP fields in mind:

1. Sender MAC Address
2. Sender IP Address
3. Target MAC Address
4. Target IP Address

These fields are important in the Dynamic ARP Inspection process.

ARP Reply

R1 sees that its IP address appears in the Target IP field and sends an ARP Reply back to PC1.

This allows PC1 to learn R1's MAC address and add it to its ARP table.

At the same time, R1 learns PC1's MAC address from the original ARP Request and creates its own ARP table entry.

Therefore, R1 does not need to send an ARP Request if it later wants to communicate with PC1.

ARP Reply Flow

- Typically it is a two message exchange: **request** and **reply**



Now PC1 knows R1's MAC address.

It can place R1's MAC address in the Ethernet frame and send the DNS query.



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Gratuitous ARP

There is another type of ARP message called **Gratuitous ARP**.

A Gratuitous ARP is:

- An ARP Reply sent without first receiving an ARP Request.
- Sent to the broadcast MAC address (all Fs).

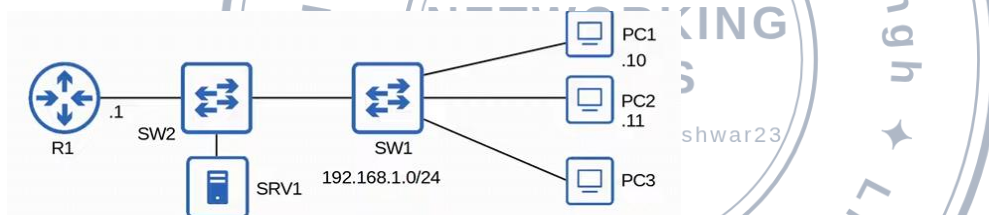
Standard ARP replies are not broadcast. They are unicast messages sent directly to the device that originated the ARP Request.

Gratuitous ARP allows devices on the local network to learn the sender's MAC address without sending ARP Requests.

Depending on the device manufacturer, Gratuitous ARP may be sent automatically when:

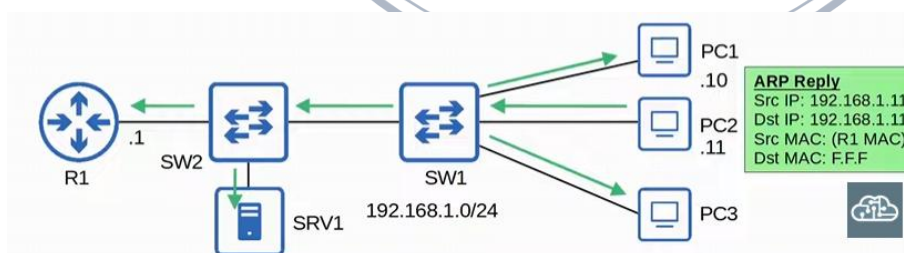
1. An interface is enabled.
2. An IP address changes.
3. A MAC address changes.

Gratuitous ARP Example:



If PC2's network interface becomes active, it may send a Gratuitous ARP Reply.

Gratuitous ARP Flow



```
ARP Reply
Src IP: 192.168.1.11
Dst IP: 192.168.1.11
Src MAC: (R1 MAC)
Dst MAC: F.F.F
```

The message is flooded throughout the local network.

As a result: Other devices add an ARP entry for PC2. Switches add entries for PC2 in their MAC address tables.

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Introduction to Dynamic ARP Inspection (DAI)

DAI is a switch feature used to filter ARP messages received on **untrusted ports**.

Important points:

1. DAI only filters ARP messages.
2. All other traffic is unaffected.
3. Like DHCP Snooping, all ports are untrusted by default.

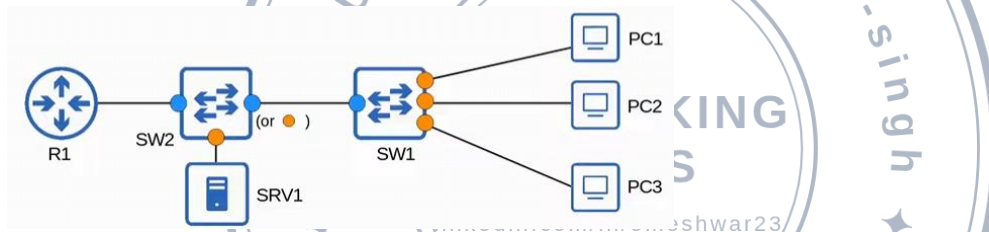
Trust Configuration

Typically:

1. Interfaces connected to switches, routers, and other network devices should be configured as **trusted**.
2. Interfaces connected to end hosts should remain **untrusted**.

Example

Blue: Trusted, Yellow: Untrusted



The SW2 interface connected to SW1 could also remain untrusted because it is a downlink toward end hosts.

However, Cisco recommends configuring interfaces connected to switches, routers, and similar devices as trusted.

How DAI Works

The operation is similar to DHCP Snooping.

Valid ARP Message

PC1 sends an ARP Request.

Process

PC1 ---> SW1 ---> SW2 ---> R1

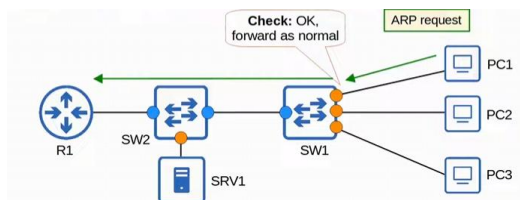
1. Message arrives on an untrusted port.
2. SW1 inspects the ARP message using DAI.
3. SW1 determines the message is valid.
4. SW1 forwards the message to SW2.

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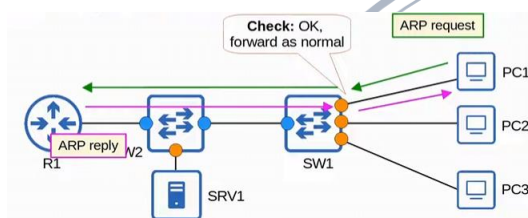
5. SW2 receives it on a trusted interface and does not inspect it.
6. SW2 forwards it to R1.
7. R1 sends the ARP Reply.
8. Neither switch inspects the reply because it arrives on trusted interfaces.

Diagram

ARP Request



ARP Reply



Invalid ARP Message

Now PC2 sends an ARP message.

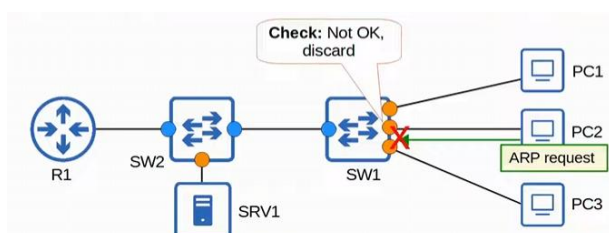
Process

PC2 ---> SW1

1. The message arrives on an untrusted port.
2. SW1 inspects it using DAI.
3. SW1 determines the message violates DAI rules.
4. SW1 discards the message.

Diagram

ARP Request



The exact rules DAI uses to determine whether a message is valid or invalid are covered later.

What Attacks Does It Prevents?

ARP Poisoning Attack

ARP poisoning is similar to DHCP poisoning.

The attacker manipulates the ARP tables of target devices so that traffic is sent to the attacker instead of the legitimate destination.

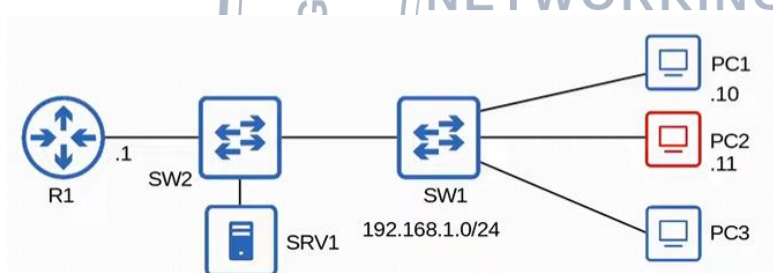
One method is to send **gratuitous ARP (GARP)** messages using another device's IP address. Another method is to send replies to legitimate ARP requests, *but this demonstration uses gratuitous ARP*.

When devices receive the gratuitous ARP message, they update their ARP tables and begin sending traffic to the attacker.

Attack Scenario

PC2 acts as the attacker.

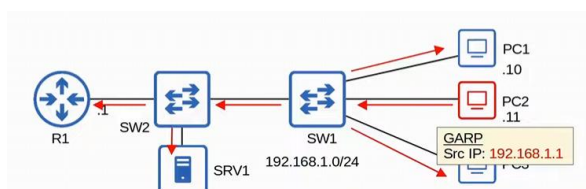
Network



Step 1: Attacker Sends Gratuitous ARP

PC2 sends a Gratuitous ARP message using the default gateway's IP address (192.168.1.1).

Gratuitous ARP Flood



The message is flooded throughout the network and all devices receive it.

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Result of the Attack

Devices update their ARP tables and associate:

192.168.1.1 ---> PC2 MAC

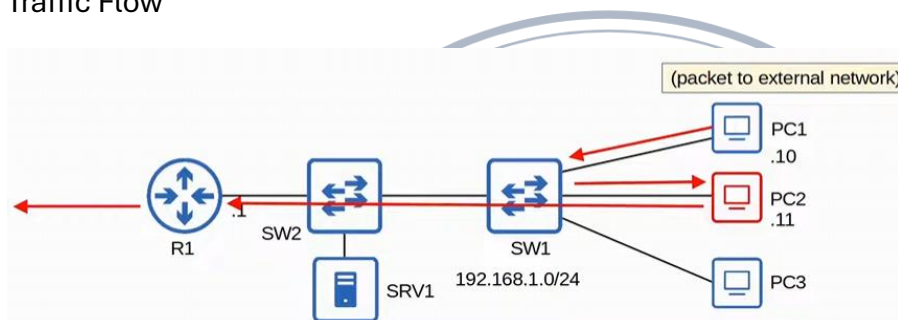
instead of associating it with R1's MAC address.

Important Note: R1 does not update its own ARP table because 192.168.1.1 is R1's own IP address.

Step 2: Traffic is Sent to the Attacker

Now when PC1 sends traffic toward an external network, it sends the frame to PC2 first.

Traffic Flow



PC2 can:

1. Save a copy of the packet for later inspection.
2. Forward the packet to the legitimate default gateway (R1).
3. Potentially manipulate the packet contents.

Man-in-the-Middle Attack

ARP poisoning can therefore be used to perform a **man-in-the-middle (MITM)** attack.

Traffic passes through the attacker before reaching its legitimate destination, similar to the DHCP poisoning attack discussed previously.

How Does It Work?

Basic Operation

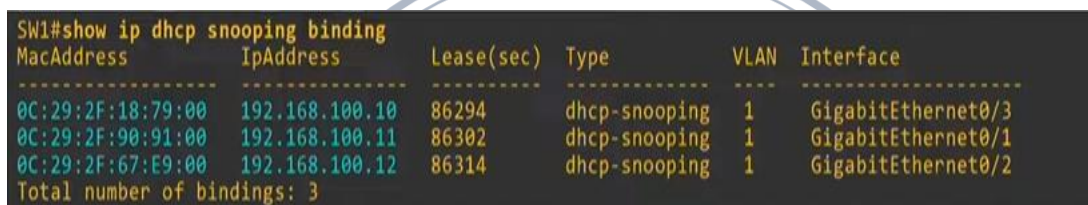
DAI examines ARP messages received on **untrusted ports**.

It checks:

1. Sender MAC Address
2. Sender IP Address

These values are compared against entries in the *DHCP Snooping Binding Table*.

DHCP Snooping Binding Table



MacAddress	IpAddress	Lease(sec)	Type	VLAN	Interface
0C:29:2F:18:79:00	192.168.100.10	86294	dhcp-snooping	1	GigabitEthernet0/3
0C:29:2F:90:91:00	192.168.100.11	86302	dhcp-snooping	1	GigabitEthernet0/1
0C:29:2F:67:E9:00	192.168.100.12	86314	dhcp-snooping	1	GigabitEthernet0/2

Total number of bindings: 3

Purpose, DHCP Snooping records the MAC and IP addresses of DHCP clients.

DAI uses this information to validate ARP messages.

Validation Process

1. If a Matching Entry Exists

DAI finds a matching IP/MAC entry in the DHCP Snooping Binding Table.

ARP Message → DAI Check → Match Found → Forward Normally

The ARP message is trusted and forwarded.

2. If No Matching Entry Exists

ARP Message → DAI Check → No Match → Discard

The ARP message is dropped.

Trusted vs Untrusted Ports

DAI only performs validation on **untrusted ports**.

Messages received on trusted ports are forwarded normally without inspection.

Like DHCP Snooping:

1. All interfaces are untrusted by default.
2. Trusted ports must be configured manually.

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ARP ACL Alternative

DAI usually relies on DHCP Snooping.

However, another option is available:

ARP ACLs

ARP ACLs can be manually configured to map:

- IP addresses
- MAC addresses

DAI can then use these mappings during validation.

This is useful for hosts that do not use DHCP.

Without DHCP:

1. No DHCP Snooping binding exists.
2. DAI would otherwise drop all ARP messages from those hosts.

ARP ACLs can be configured for specific hosts, or for all hosts if desired.

Additional DAI Features

Besides checking:

1. Sender MAC Address
2. Sender IP Address

DAI can also perform additional, more detailed checks.

These checks are optional.

DAI Rate Limiting

Like DHCP Snooping, DAI supports rate limiting.

Both DHCP Snooping and DAI require CPU processing on the switch.

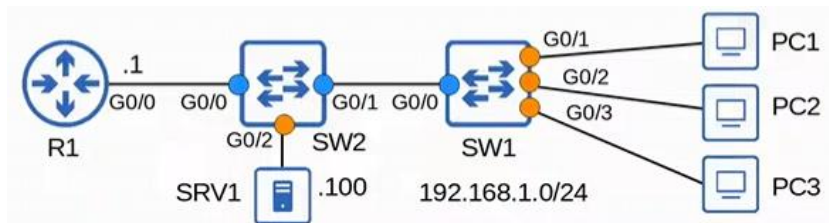
Even if malicious ARP messages are blocked, an attacker could still attempt to overload the switch CPU by sending a large number of ARP messages.

If this occurs, rate limiting can disable the interface.

This provides protection against such attacks.

DAI Configuration

Configuration of Dynamic ARP Inspection



Enable DAI on a VLAN

```
SW2(config)#ip arp inspection vlan 1
```

In this network, VLAN 1 is used.

If multiple VLANs exist, DAI should be enabled on each VLAN.

Otherwise, only ARP messages belonging to the configured VLAN will be inspected.

Configure Trusted Ports

```
SW2(config)#ip arp inspection vlan 1
SW2(config)#interface range g0/0 - 1
SW2(config-if-range)#ip arp inspection trust
```

- DAI enabled on VLAN 1.
- G0/0 and G0/1 configured as trusted.

SW1

```
SW1(config)#ip arp inspection vlan 1
SW1(config)#interface g0/0
SW1(config-if)#ip arp inspection trust
```

- DAI enabled on VLAN 1.
- Only G0/0 configured as trusted.

These are the basic commands required to enable DAI and configure trusted interfaces.

DAI vs DHCP Snooping Enable Commands

```
DHCP snooping requires two commands to enable it:
ip dhcp snooping
ip dhcp snooping vlan vlan-number
DAI only requires one:
ip arp inspection vlan vlan-number
```

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Show Command

Display DAI Interface Information

SW1#show ip arp inspection interfaces

Interface	Trust State	Rate (pps)	Burst Interval
Gi0/0	Trusted	None	N/A
Gi0/1	Untrusted	15	1
Gi0/2	Untrusted	15	1
Gi0/3	Untrusted	15	1
Gi1/0	Untrusted	15	1
Gi1/1	Untrusted	15	1
Gi1/2	Untrusted	15	1
Gi1/3	Untrusted	15	1
Gi2/0	Untrusted	15	1
Gi2/1	Untrusted	15	1
Gi2/2	Untrusted	15	1
Gi2/3	Untrusted	15	1
Gi3/0	Untrusted	15	1
Gi3/1	Untrusted	15	1
Gi3/2	Untrusted	15	1
Gi3/3	Untrusted	15	1

DAI rate limiting is enabled on untrusted ports by default with a rate of 15 packets per second. It is disabled on trusted ports by default. *DHCP snooping rate limiting is disabled on all interfaces by default.

DHCP snooping rate limiting is configured like this:
x packets per second.

The DAI **burst interval** allows you to configure rate limiting like this:
x packets per y seconds

Trust State-Shows whether an interface is:

- Trusted
- Untrusted

On SW1, only **Gi0/0** is trusted.

Rate Limiting Column-Shows DAI rate-limiting settings.

Default Behaviour

DAI

1. Enabled on untrusted ports by default.
2. Default rate: 15 packets per second.
3. Disabled on trusted ports by default.

DHCP Snooping

1. Rate limiting disabled on all interfaces by default.
2. Applies to both trusted and untrusted ports.

Burst Interval

DAI includes a burst interval feature.

DHCP Snooping Rate Limit- X packets per second

DAI Rate Limit- X packets per Y seconds

The measured interval does not have to be one second.

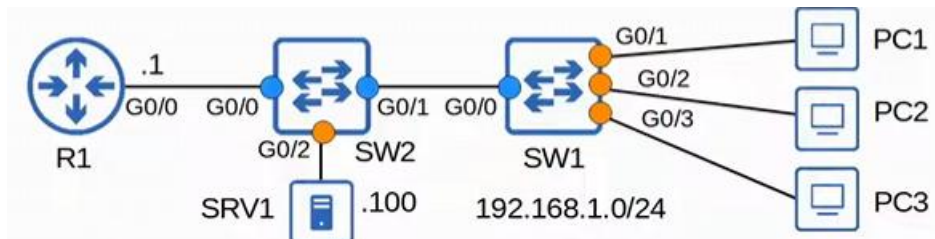
This provides greater flexibility when controlling the rate of ARP messages on an interface.

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DAI Rate Limiting Configuration

Previously, we saw that DAI rate limiting is enabled by default on untrusted ports with a rate of 15 packets per second. Now let's see how to configure it.

Network Topology



Configuring Rate Limiting

```
SW1(config)#interface range g0/1 - 2
SW1(config-if-range)#ip arp inspection limit rate 25 burst interval 2
SW1(config-if-range)#interface range g0/3
SW1(config-if)#ip arp inspection limit rate 10
SW1(config-if)#do show ip arp inspection interfaces
```

The burst interval is optional. If you don't specify it, the default is 1 second.

Interface	Trust State	Rate (pps)	Burst Interval
Gi0/0	Trusted	None	N/A
Gi0/1	Untrusted	25	2
Gi0/2	Untrusted	25	2
Gi0/3	Untrusted	10	1

![output omitted]

If ARP messages are received faster than the specified rate, the interface will be err-disabled. It can be re-enabled in two ways:
1: **shutdown** and **no shutdown**
2: **errdisable recovery cause arp-inspection**

Explanation

For interfaces G0/1 and G0/2: ip arp inspection limit rate 25 burst interval 2

means: 25 ARP packets per 2 seconds

For interface G0/3: ip arp inspection limit rate 10

means: 10 ARP packets per second, because no burst interval was specified.

Burst Interval:

The burst interval is optional.

If it is not configured, the default value is: 1 second

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What Happens When the Limit is Exceeded?

If ARP messages are received faster than the configured rate, the interface is placed into: err-disabled state.

Important:

1. Rate limiting applies to ARP messages **received** on an interface.
2. It does not limit ARP messages **sent** by an interface.

Recovering an Err-Disabled Interface

An interface disabled by DAI can be re-enabled in two ways:

1. Manually

1. shutdown
2. no shutdown

2. Errdisable Recovery

```
SW1(config)#errdisable recovery cause arp-inspection
SW1(config)#do show errdisable recovery
ErrDisable Reason    Timer Status
-----
arp-inspection       Enabled
![outputted omitted]
```

This enables automatic recovery for interfaces disabled by ARP inspection.

Additional DAI Validation Checks

By default, DAI verifies:

1. Sender MAC Address
2. Sender IP Address

and checks for a matching entry in the DHCP Snooping Binding Table.

Additional validation checks can be enabled with: ip arp inspection validate

Available options:

```
SW1(config)#ip arp inspection validate ?
dst-mac  Validate destination MAC address
ip       Validate IP addresses
src-mac  Validate source MAC address
```

dst-mac: Enables validation of the destination MAC address in the Ethernet header against the target MAC address in the ARP body for ARP responses. The device classifies packets with different MAC addresses as invalid and drops them

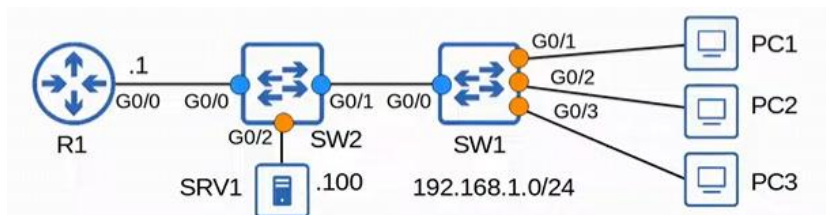
ip: Enables validation of the ARP body for invalid and unexpected IP addresses. Addresses include 0.0.0.0, 255.255.255.255, and all IP multicast addresses. The device checks the sender IP addresses in all ARP requests and responses and checks the target IP addresses only in ARP responses.

src-mac: Enables validation of the source MAC address in the Ethernet header against the sender MAC address in the ARP body for ARP requests and responses. The device classifies packets with different MAC addresses as invalid and drops them.

(source: https://www.cisco.com/c/en_us/techdoc/dc/reference/cli/h5k/commands/ip-arp-inspection-validate.html)

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Configuring Multiple Validation Checks



```
SW1(config)#ip arp inspection validate dst-mac
SW1(config)#ip arp inspection validate ip
SW1(config)#ip arp inspection validate src-mac

SW1(config)#do show running-config | include validate
ip arp inspection validate src-mac
```

Only the last command appears.

Why?

Each new command overwrites the previous one.

Correct Method

To enable multiple checks, configure them together:

```
SW1(config)#ip arp inspection validate ip src-mac dst-mac
SW1(config)#do show running-config | include validate
ip arp inspection validate src-mac dst-mac ip
```

You must enter all of the validation checks you want in a single command.
*You can specify one, two, or all three.
*The order isn't significant.

Summary

All desired validation checks must be entered in a single command.

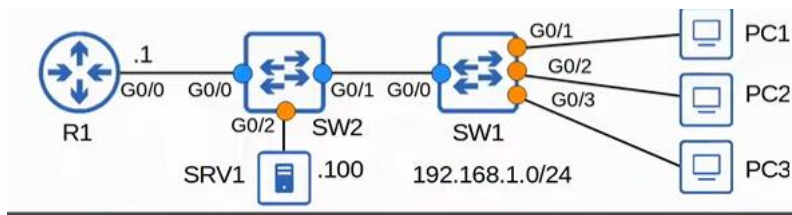
You may specify:

1. One check
2. Two checks
3. All three checks

The order is not significant.

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ARP ACLs



ARP ACLs provide another method for DAI validation.

This is useful for devices that do not use DHCP.

Problem Scenario

DHCP Snooping Binding Table

```
SW2#show ip dhcp snooping binding
MacAddress      IpAddress      Lease(sec)  Type           VLAN  Interface
-----
0C:29:2F:18:79:00 192.168.1.12   79226      dhcp-snooping  1     GigabitEthernet0/1
0C:29:2F:90:91:00 192.168.1.10   79188      dhcp-snooping  1     GigabitEthernet0/1
0C:29:2F:67:E9:00 192.168.1.11   79210      dhcp-snooping  1     GigabitEthernet0/1
Total number of bindings: 3

!SRV1 has a static IP address of 192.168.1.100, so it does not have an entry in SW2's DHCP
!snooping binding table.
```

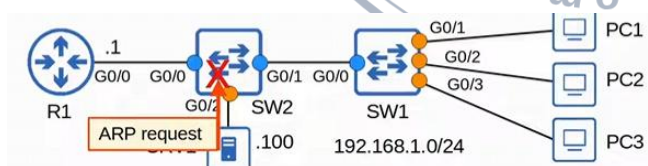
SRV1 uses a static IP address: 192.168.1.100

Therefore: No DHCP Snooping entry exists for SRV1.

Result

When SRV1 sends an ARP request, DAI drops it.

```
*Jun 19 05:56:15.538: %SW DAI-4-DHCP_SNOOPING_DENY: 1 Invalid ARPs (Req) on Gi0/2, vlan 1.
([0c29.2f1e.7700/192.168.1.100/0000.0000.0000/192.168.1.1/05:56:14 UTC Sat Jun 19 2021])
```



Creating an ARP ACL

```
SW2(config)#arp access-list ARP-ACL-1
SW2(config-arp-nacl)#permit ip host 192.168.1.100 mac host 0c29.2f1e.7700
```

This creates an ARP ACL permitting SRV1.

Applying the ARP ACL

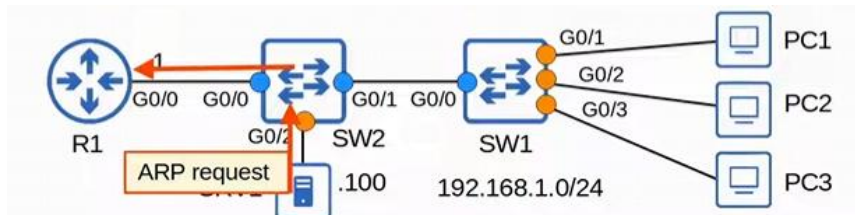
Creating the ACL alone is not enough, It must be applied.

```
SW2(config)#ip arp inspection filter ARP-ACL-1 vlan 1
```

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Result

Now when SRV1 sends an ARP request:



The ARP message is forwarded even though SRV1 does not have an entry in the DHCP Snooping Binding Table.

The ARP ACL permits the traffic.

Show Command

Display DAI Information

```
SW2#show ip arp inspection
```

Vlan	Configuration	Operation	ACL Match	Static ACL
1	Enabled	Active	ARP-ACL-1	No

Vlan	ACL Logging	DHCP Logging	Probe Logging
1	Deny	Deny	Off

Vlan	Forwarded	Dropped	DHCP Drops	ACL Drops
1	56	4	4	0

Vlan	DHCP Permits	ACL Permits	Probe Permits	Source MAC Failures
1	0	1	0	0

Vlan	Dest MAC Failures	IP Validation Failures	Invalid Protocol Data
1	0	0	0

- If **static ACL** is set to **yes**, the implicit deny at the end of the ARP ACL will take effect.
- This will cause all ARP messages not permitted by the ARP ACL to be denied.
- In effect, this means that only the ARP ACL will be checked, the DHCP snooping table will not be checked.

Command Summary

```
SW1(config)# ip arp inspection vlan vlan-number
SW1(config)# errdisable recovery cause arp-inspection
SW1(config)# ip arp inspection validate (src-mac | dst-mac | ip)
SW1(config-if)# ip arp inspection trust
SW1(config-if)# ip arp inspection limit rate packets [burst interval seconds]

SW1(config)# arp access-list name
SW1(config-arp-nacl)# permit ip host ip-address mac host mac-address
SW1(config)# ip arp inspection filter arp-acl-name vlan vlan-number

SW1# show ip arp inspection
SW1# show ip arp inspection interfaces
```

